

Practical 2

Name : Prajwal Dudhal

Roll No : 13154

Batch: A3

```
import time
```

```
import random
```

```
# Thresholds
```

```
TEMP_THRESHOLD = 28 # Celsius
```

```
HUMID_THRESHOLD = 65 # Percentage
```

```
def read_sensor():
```

```
    # Simulate DHT11 sensor readings
```

```
    temperature = round(random.uniform(24, 35), 1)
```

```
    humidity = round(random.uniform(55, 80), 1)
```

```
    return humidity, temperature
```

```
print("Temperature & Humidity Monitor Simulation")
```

```
print(f"Thresholds — Temp: {TEMP_THRESHOLD}°C | Humidity: {HUMID_THRESHOLD}%")
```

```
print("Press Ctrl+C to stop.\n")
```

```
try:
```

```
    while True:
```

```
        humid, temp = read_sensor()
```

```
        print(f"Humidity: {humid}% | Temperature: {temp}°C")
```

```
        if humid > HUMID_THRESHOLD:
```

```
            print(" → HUMIDITY HIGH [HUMIDITY LED ON]")
```

```

else:

    print(" → Humidity Normal [HUMIDITY LED OFF]")

if temp > TEMP_THRESHOLD:

    print(" → TEMP HIGH    [TEMP LED ON] ⚠ ALERT!")
else:

    print(" → Temp Normal    [TEMP LED OFF]")

print()

time.sleep(1)

except KeyboardInterrupt:

    print("Simulation stopped.")

```

Output

Temperature & Humidity Monitor Simulation

Thresholds — Temp: 28°C | Humidity: 65%

Press Ctrl+C to stop.

Humidity: 65.7% | Temperature: 30.3°C

→ HUMIDITY HIGH [HUMIDITY LED ON]

→ TEMP HIGH [TEMP LED ON] ⚠ ALERT!

Humidity: 71.6% | Temperature: 28.6°C

→ HUMIDITY HIGH [HUMIDITY LED ON]

→ TEMP HIGH [TEMP LED ON] ⚠ ALERT!